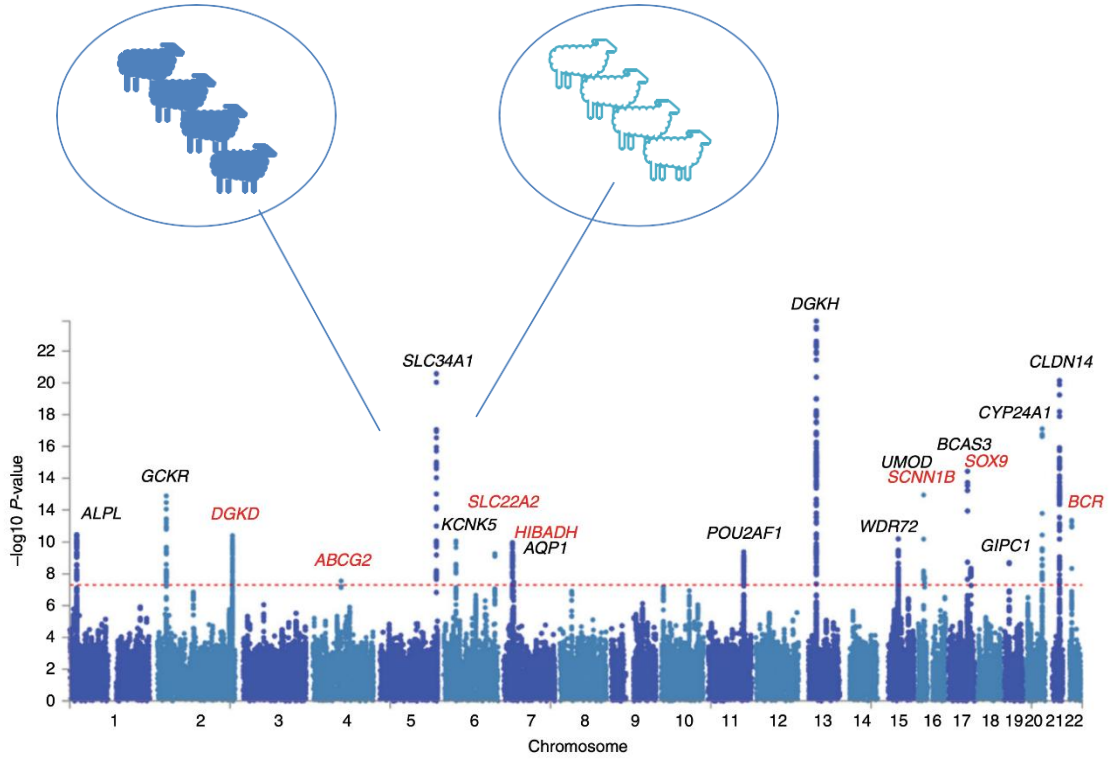
GWAS Analysis for Behavioral Traits in Domestic Sheep



Training Material
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20.04.2026

Introduction to GWAS (Genome-Wide Association Studies)

What is GWAS?

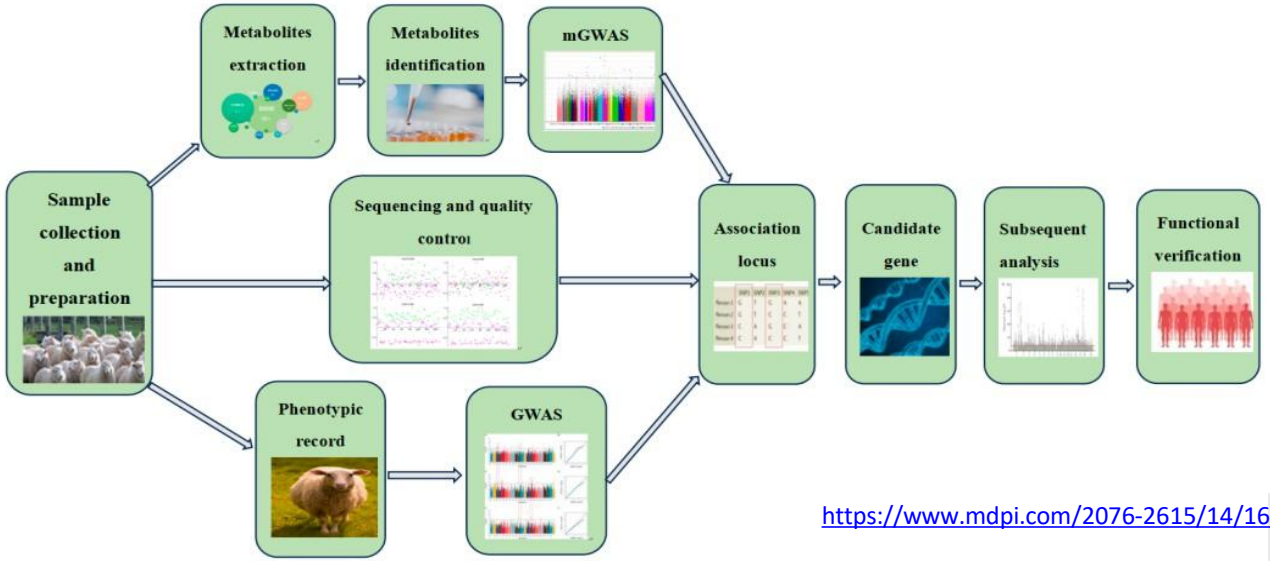
A statistical approach testing associations between genetic variants and phenotypes across genome

Detection of differential allele frequencies of variants to reveal complex trait variation (Uffelmann, 2021)

The associated variants may be the causal or linked to the causal ones due to linkage disequilibrium

They partially control phenotypic variation

www.genome.gov/genetics-glossary/Single-Nucleotide-Polymorphisms-SNPs



https://www.mdpi.com/2076-2615/14/16/2382

Applications in Livestock Genomics

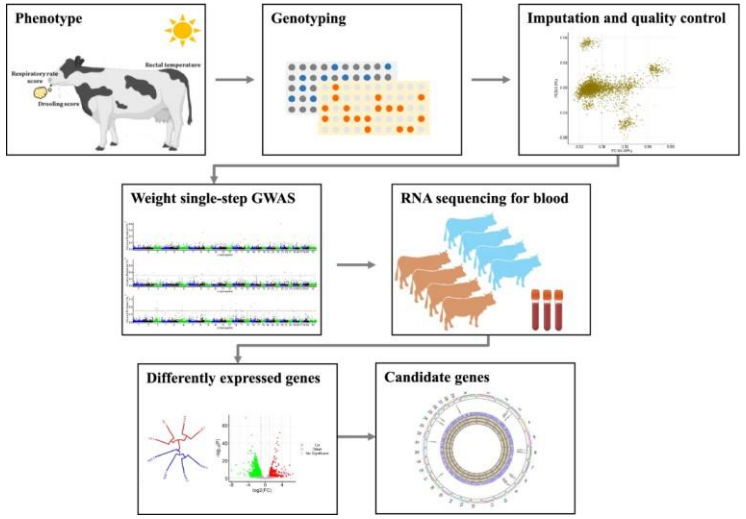
Traits of economic interest: production, reproduction, health, longevity

Quantitative characters - continuous variation within/between populations

Influenced by multiple genes with relative effects - polygenic inheritance

GWAS enable the detection of specific SNP markers

Implementation of genomic selection for trait improvement



https://jasbsci.biomedcentral.com/articles/10.1186/s40104-022-00748-6

Increasing interest on adaptive and behavioral traits

Understanding of local adaptation, animal resilience/behavior to environmental changes

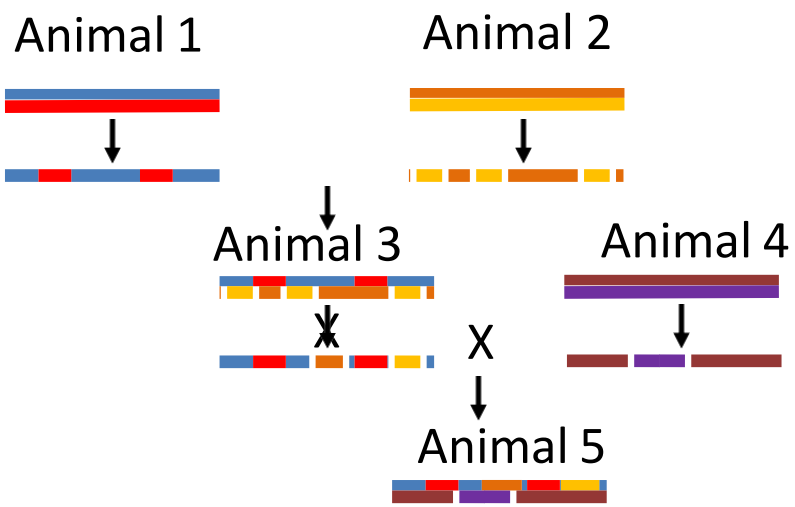
What does a phenotype consist of?

Phenotype (P) = Genotype (G) + Environment (E)

- **Phenotype** : Trait, observable characteristic of an animal
- **Genotype** : Genetic profile of an individual. We can act on it. Inherited
- **Environment** : Events around the individual. We can partially act on it. Not inherited

What does a genotype represent?

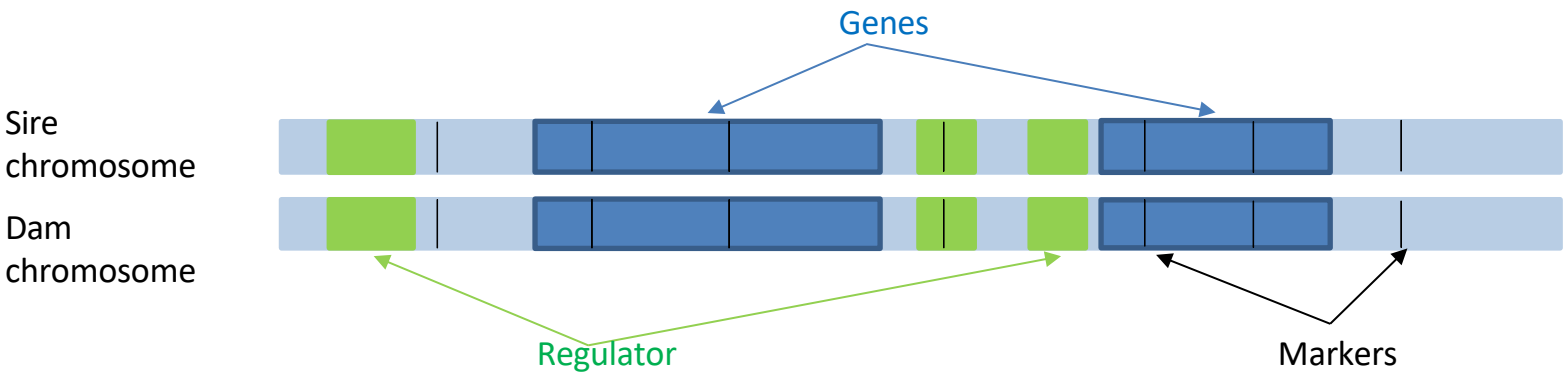
- **Genotype:** Determined by alleles/DNA segments across genome of an individual coding for functions related to the trait (also used to describe the alleles on a particular locus)
- **Haplotype:** A group of tightly linked alleles/DNA segments from a single parent inherited together
- **Haplotype Blocks:** haplotypes of different length are sometimes co-inherited (Linkage Disequilibrium)



- **Recombination reduces LD (shorter haplotypes)**
- **Selection increase LD (longer haplotypes)**

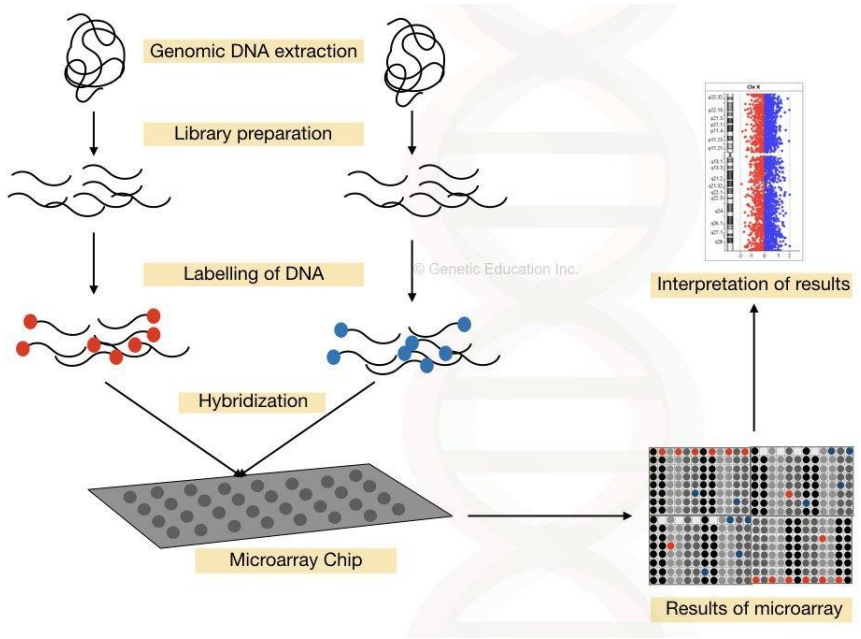
What is a genetic marker?

- **Markers: Variable sites/sequences with known position on the genome**



- **They can be linked with genes related to a trait of interest**
- **If they are linked with relevant genes, we can use them for selection purposes**
- **We can also use them to estimate the relationships between individuals**

High-throughput genotyping and GWAS



https://geneticeducation.co.in/snp-array-high-throughput-snp-genotyping-technique/

SNP array

DNA microarray enabling genotyping of thousands to millions of SNPs to detect variations across the genome simultaneously

Key features

Species-specific design (e.g., livestock arrays)

Cover the genome at medium or high density

Allow large-scale population analysis

GWAS studies

Testing associations between every marker included in the array with the phenotype of interest

Animal Model

Statistical methods first developed by Fisher (1918) and Wright (1921) to **partition the observed phenotypic variation into variance components (genetic, environmental)**

Phenotype (P) = Genotype (G) + Environment (E)

$$Y_i = X + u_i + \varepsilon$$

- **Phenotype (Y)** : Trait, observable characteristic of an individual
- **Fixed effect (X)** : Part of the environment clearly identified, eg: Farm, sex, year, …
- **Random effect (u)** : Animal genetic profile – estimated with pedigree information
- **Error (ε):** Residual due to imperfection of the model

GWAS Model

Test the association of each SNP marker with the phenotype

What is new?

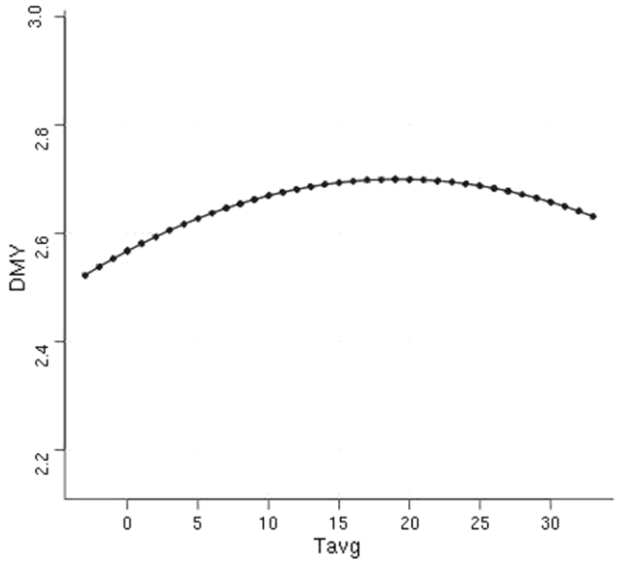
GWAS model ➔ ***Y*** = ***X*** + ***Marker effect*** + ***u*** + ***ε***

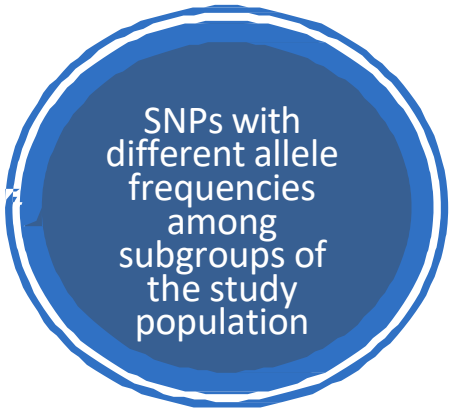
- **Phenotype (Y)** : Trait, observable characteristic of an animal
- **Fixed effect (X)** : Environmental factors clearly identified, eg: Farm, sex, year, …
- **Marker effect**: each SNP is tested individually
- **Random effect (u)** : random (additive genetic) polygenic effect → Genomic Relatedness Matrix (GRM)
- **Error (ε):** Residual due to imperfection of the model

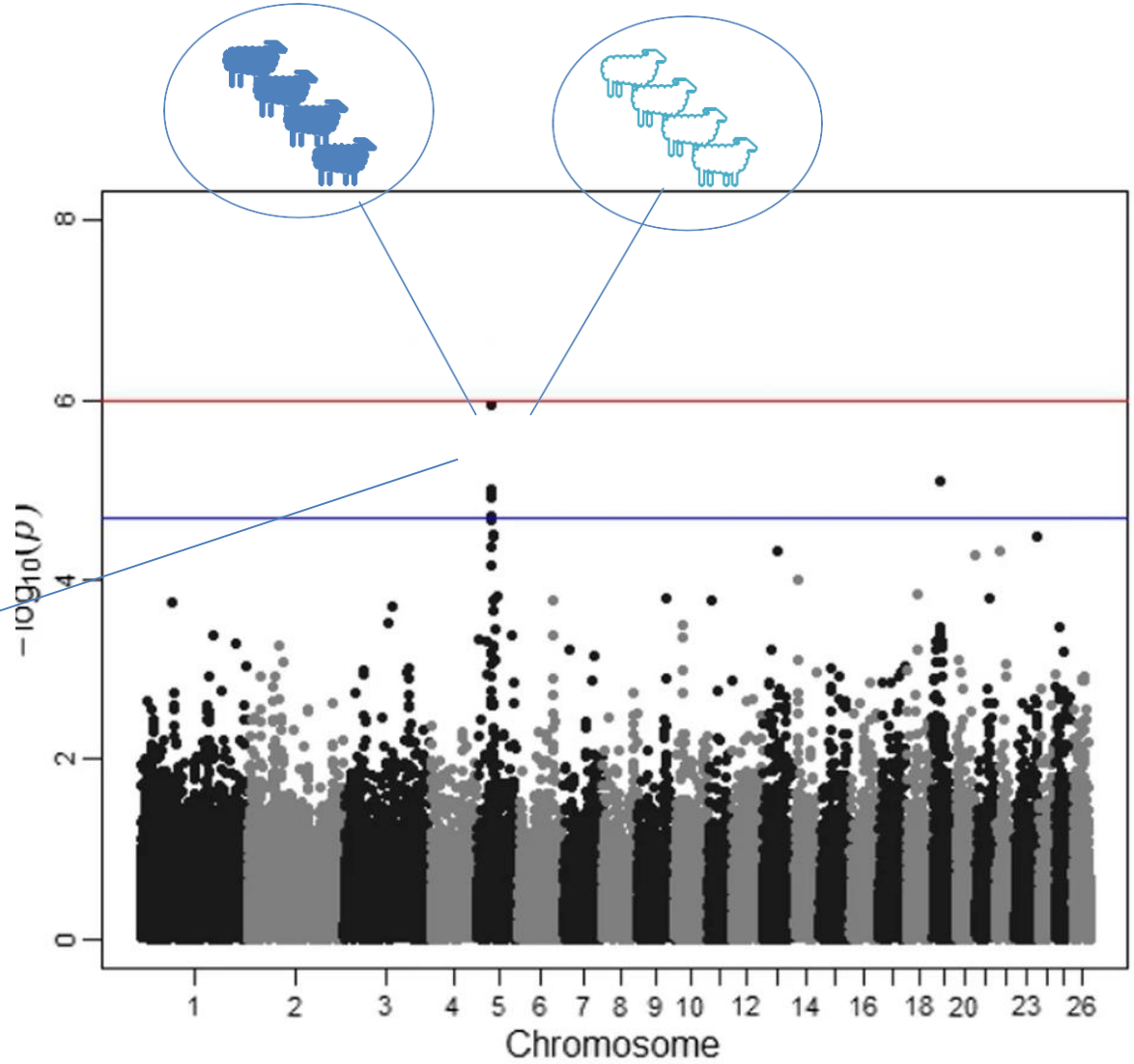
↓

Control for genetic relatedness between individuals that may lead to false positive associations

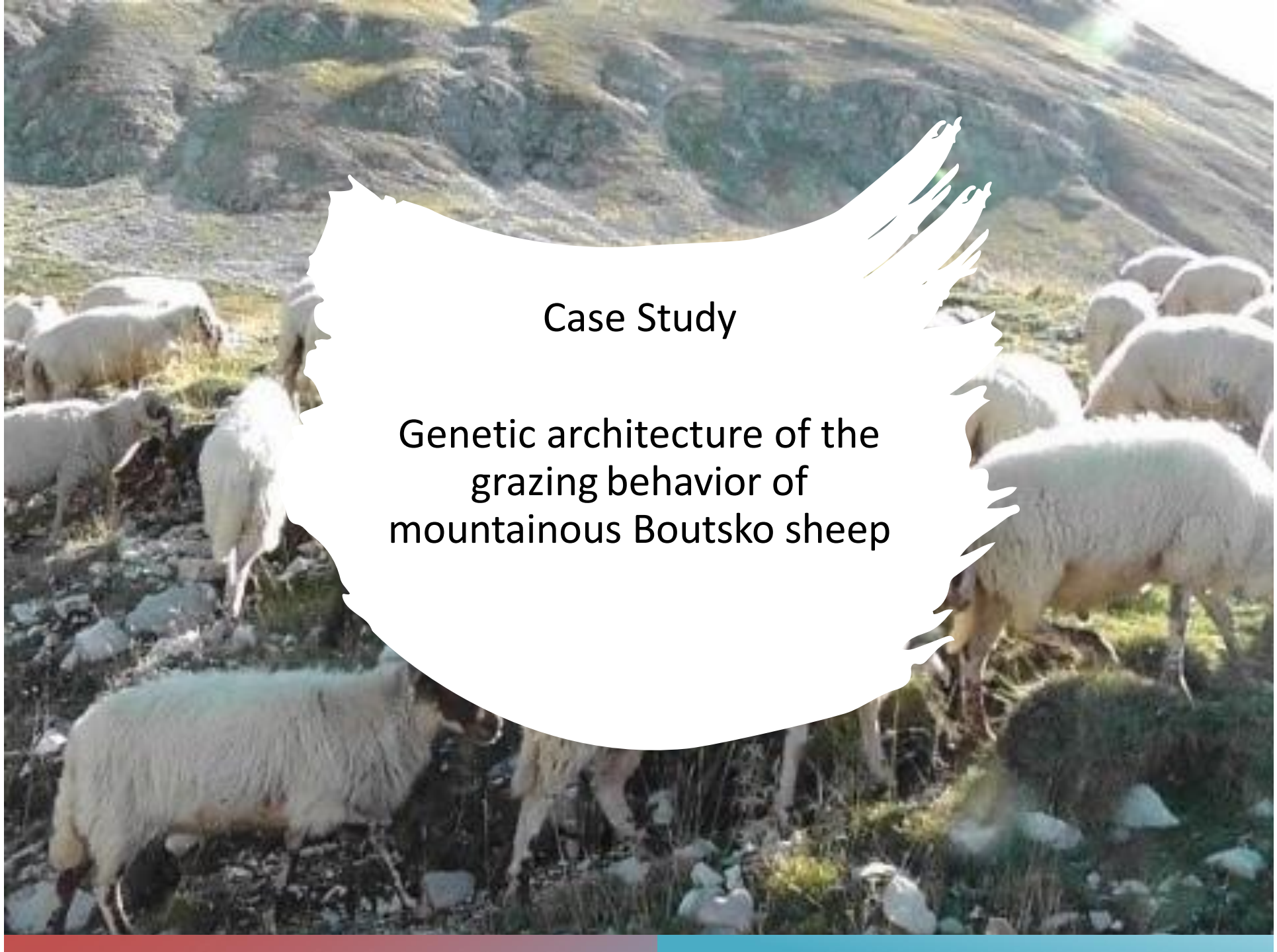
Example of GWAS results for climate resilience of Greek Chios sheep







https://link.springer.com/article/10.1186/s12711-021-00682-7



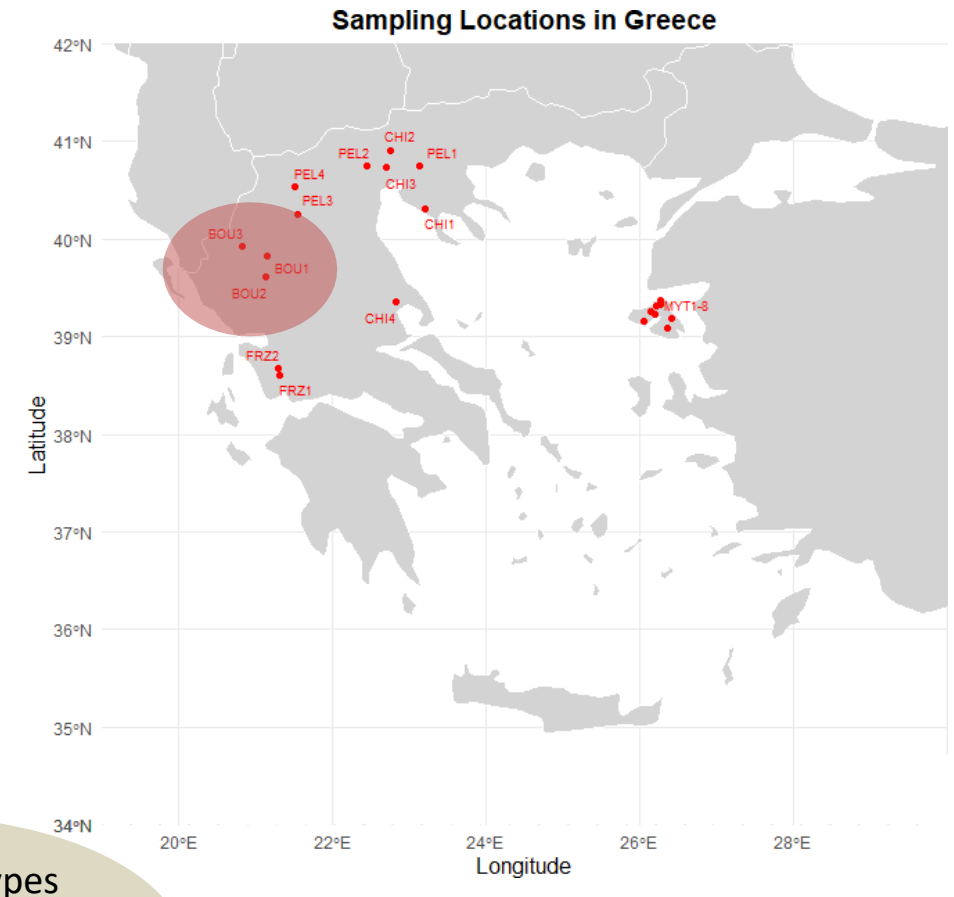
Case Study

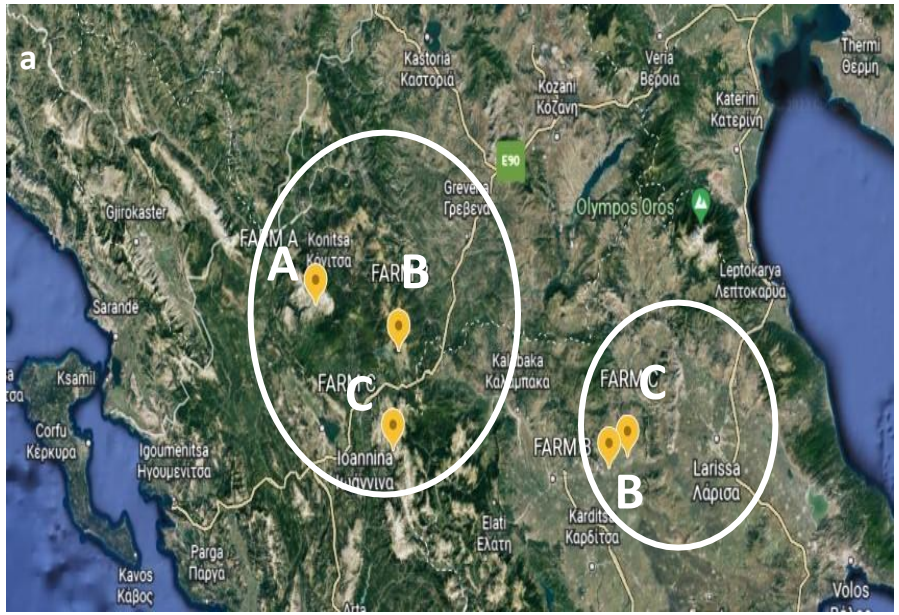
Genetic architecture of the grazing behavior of mountainous Boutsko sheep

Study Design – Overview

- Phenotype collection
- Blood sampling
- DNA extraction and SNP Genotyping
- GWAS Workflow

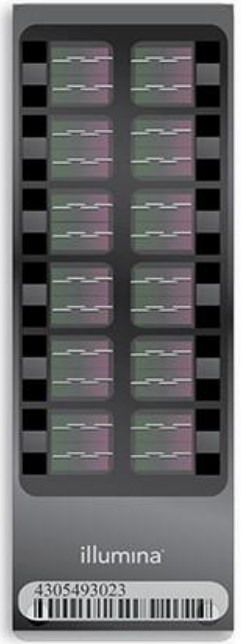
Sampling locations & Animal monitoring





Studied phenotypes
Daily grazing duration
Altitude difference

Smarter
SMAll RuminanTs breeding for Efficiency and Resilience



SNP Genotyping across genome

- Illumina Ovine SNP50 BeadChip – 53,516 SNPs
  (https://www.illumina.com/documents/products/datasheets/datas heet_ovinesnp50.pdf)

Ped file

```
0 1_A01 0 0 0 -9 A G G G A A C C A G C C G G G G A A A A A G G G
0 1_B01 0 0 0 -9 A A A G A G C C A A A A A G A G A G A A A G G G
0 1_C01 0 0 0 -9 A G A G A A C C A G A C G G G G G G A A A G G G
0 1_D01 0 0 0 -9 A A A A A A A A A A A A A A A A A A A A A A A A
0 1_E01 0 0 0 -9 A A G G A G C C A A A C G G A G G G A A A A A G
0 1_F01 0 0 0 -9 A A A G A G C C A G C C A G A A G G A A A A A A
0 1_G01 0 0 0 -9 G G A G A A C C A A A C A G G G G G A A A G A G
0 1_H01 0 0 0 -9 A G A G A A C C A G C C G G A A A G A A A G A G
0 1_A02 0 0 0 -9 A A A G A A C C A G A C G G A G G G A A A G A G
0 1_B02 0 0 0 -9 A A A G A G A C G G C C A G G G G G A A A A A G
0 1_C02 0 0 0 -9 A G A G A A C C A A C C A A A G A A A A A G A G
0 1_D02 0 0 0 -9 A A A A A A C C A G A C G G A A G G A A G G A A
0 1_E02 0 0 0 -9 A G A G A A C C A G A C A G A G A G A A G G G G
0 1_F02 0 0 0 -9 A A G G A A C C A G C C A G G G G G A A A G G G
```

map file

```
23      250506CS3900140500001_312.1     0       26243215
7       250506CS3900176800001_906.1     0       81590897
16      250506CS3900211600001_1041.1    0       41363310
2       250506CS3900218700001_1294.1    0       148834939
1       250506CS3900283200001_442.1     0       188328803
11      250506CS3900371000001_1255.1    0       35291132
16      250506CS3900386000001_696.1     0       62648296
1       250506CS3900414400001_1178.1    0       103285485
12      250506CS3900435700001_1658.1    0       45150716
1       250506CS3900464100001_519.1     0       85655170
14      250506CS3900487100001_1521.1    0       1100680
27      250506CS3900539000001_471.1     0       115555687
2       250506CS3901012300001_913.1     0       100485014
7       250506CS3901300500001_1084.1    0       89370728
3       CL635241_413.1  0       181982922
3       CL635750_128.1  0       223451754
6       CL635944_160.1  0       114661604
6       Contig35697_5761.1      0       18784742
0       CR_594.1        0       0
27      CR_816.1        0       56338477
28      CytB_1406.1     0       14554
28      CytB_1505.1     0       14653
```

binary format

.fam/.bed/.bim

GWAS Workflow

1. Genotype data quality control

2. Assessment of population stratification

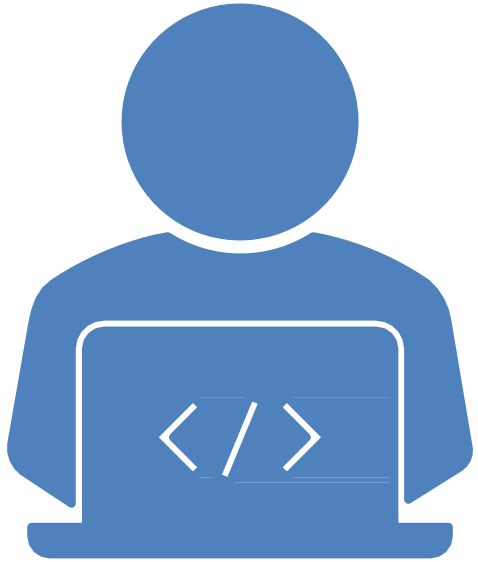
in our case, principal components explain the farm structure, so the farm is fitted as effect in the model

3. Association testing (GWAS)

4. Significance testing of raw p-values with Bonferroni correction method

https://onlinelibrary.wiley.com/doi/epdf/10.1002/mpr.1608

Software and Tools



- PLINK v1.9 https://hpc.it.auth.gr/applications/plink/
- GEMMA https://hpc.it.auth.gr/applications/gemma/
- RStudio

Quality Control

Standard filtering steps using plink

- Exclude non-autosomal and unmapped SNPs
- SNP call rate
- Minor allele frequency
- Individual call rate

Genome-wide Association Study (GWAS)

GWAS with GEMMA

Linear mixed model

Continuous trait

Construction of GRM → Matrix expressing the relationships among the individuals

Covariate File (fixed effects and covariates → A row per animal in the same order of the geno file that includes all env factors significantly affecting phenotype variation

Farm
Distance
Speed
elevation gain

Filtered genotypes with phenotype information

Quality Control implementation - PLINK: Whole genome data analysis toolset

1. **Load plink from** https://hpc.it.auth.gr/applications/plink/

2. Remove individuals with no available phenotypes

```
</>  Bash

plink --bfile boutsko3 --chr-set 26 –allow-extra-chr --remove boutsko_no-pheno.txt --make-bed --out boutsko_only-pheno
```

3. Exclude unmapped SNP markers

```
</>  Bash
input file= previous file
use snps_bou_remove.txt to exclude unmapped snps
create new binary file for next step
```

Plink v1.9 tutorial

https://www.cog-genomics.org/plink/

4. Perform quality control

```
</>  Bash
input file= previous file
Set the snp call rate=0.02, maf=0.02, and sample call rate=0.05
create new binary file for next step
```

https://github.com/valentinatsar/GWAS/tree/main

Use the link to access the material for GWAS workflow with a dataset of 292 individuals featuring 51,725 SNPs per genotype

GWAS results – genome-wide level
Visualization in RStudio

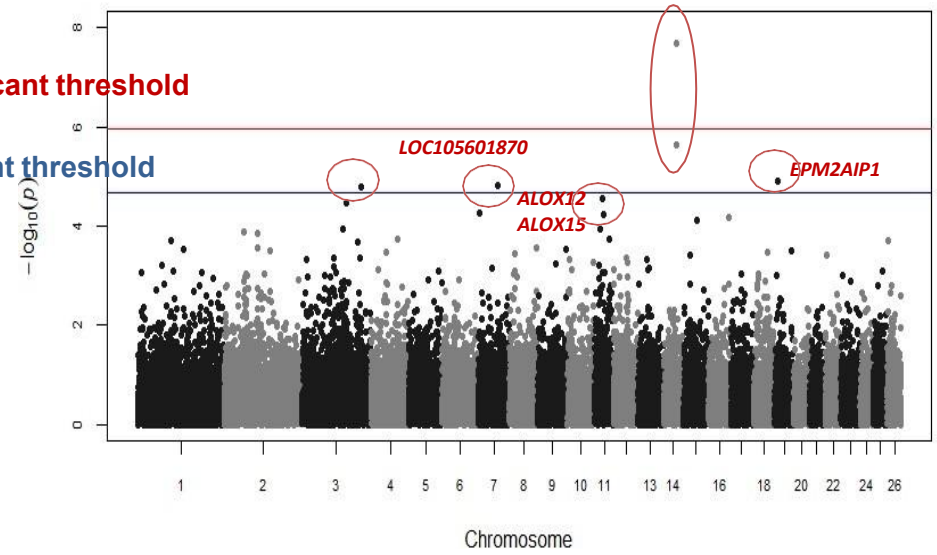
Manhattan plots

Association between SNPs and phenotype under study

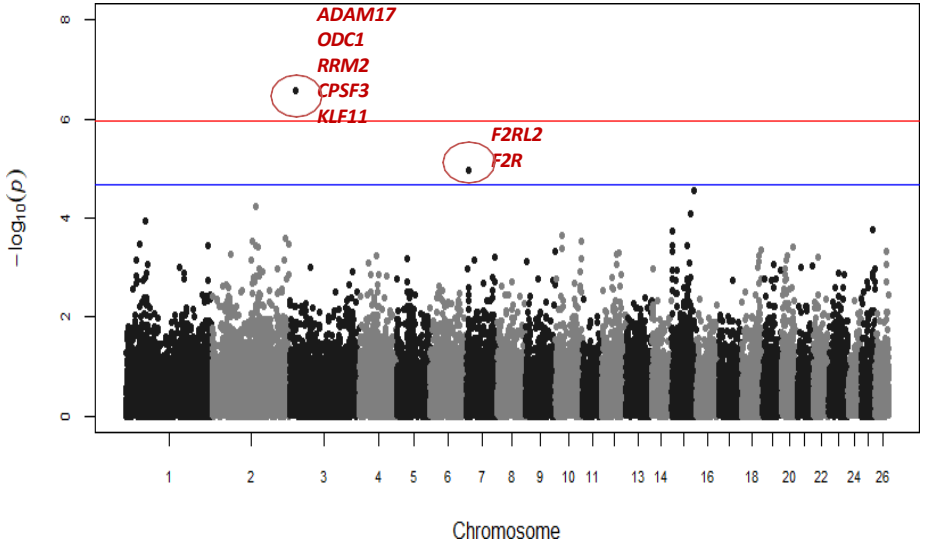
Daily grazing duration

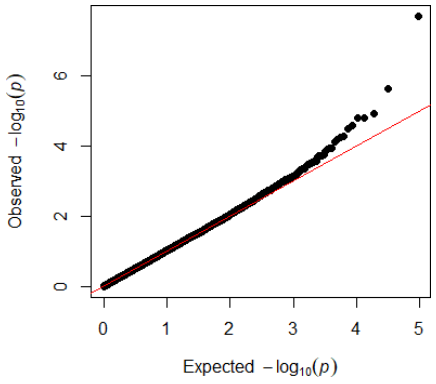
Genome-wide significant threshold

suggestive significant threshold



Altitude difference

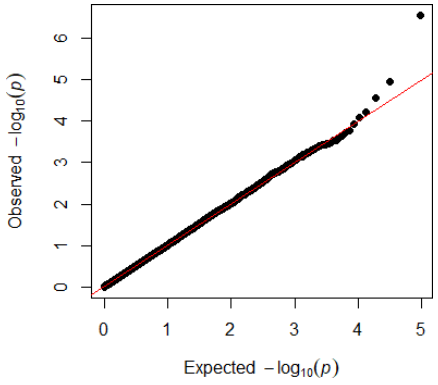




QQ plots

Expected vs Observed p-values

Early deviations from diagonal line suggests hidden population structure



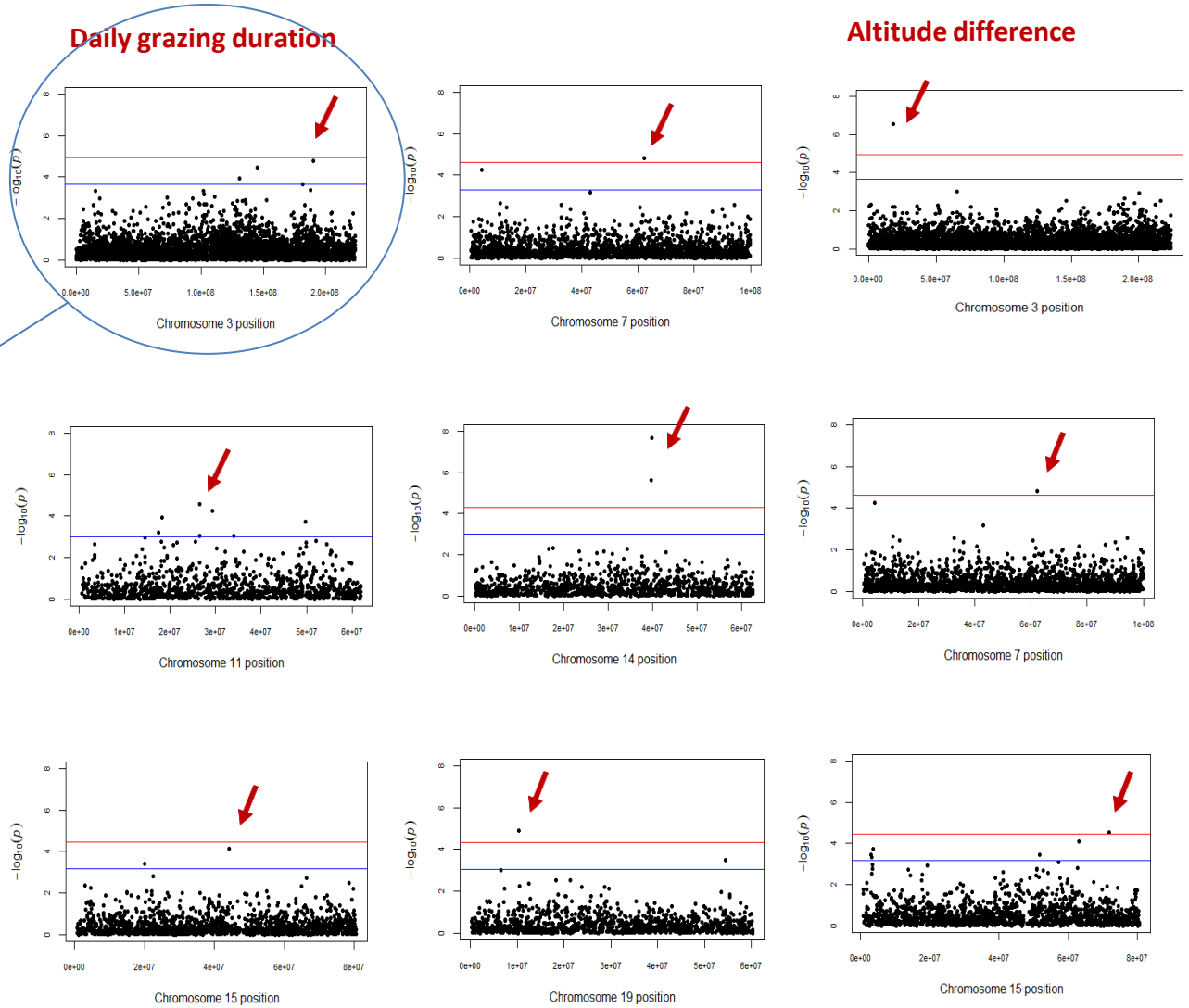
GWAS results – chromosome-wide level Visualization in RStudio

Daily grazing duration

Altitude difference

Genome-wide significant threshold

suggestive significant threshold

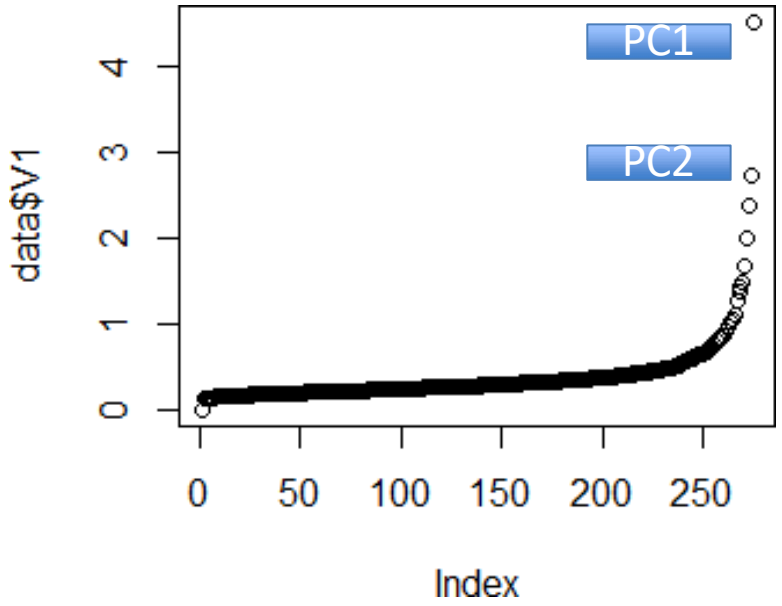


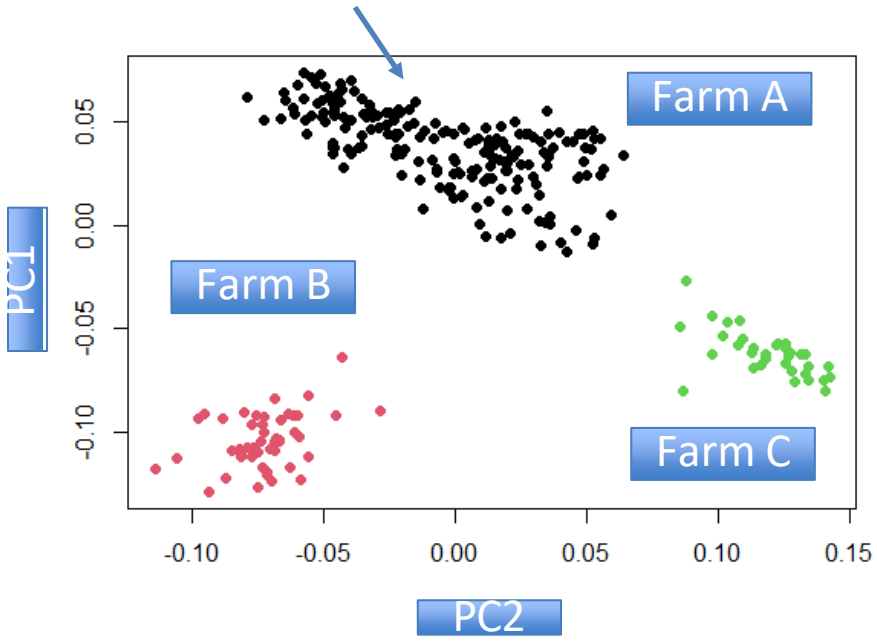
Suggestive significant SNPs on chr3 will be annotated in lecture 2

PCA results – Visualization in Rstudio

optional step – code include in github repository to decompose GRM in principal components

principal components are usually included in GWAS as covariates to correct for population structure
our case: farm=population structure





eigen values = variance explained by each PC

eigen vectors = principal components = directions of data variation